## Part 1 - Lab Objectives (Chapter 2)

- LO 2.1 Data Management/Wrangling.
- LO 2.2 Data Inspection.
- LO 2.3 Data Preparation.
- LO 2.4 Transforming Numerical Data.
- LO 2.5 Transforming Categorical Data.

## Exercises

**Question 1 (2.1)**. Which of the following statements correctly describe the data wrangling process? Select all that apply.

a. Data wrangling is the process of retrieving, cleansing, integrating, transforming, and enriching data.
b. Data wrangling is the process of defining and modeling the structure of a database to represent real-world events.
c. The objectives of data wrangling include improving data quality and reducing the time and effort required to perform analytics.
d. Data wrangling focuses on transforming the raw data into a format that is more appropriate and easier to analyze.

**Answer 1.** Statements 1, 3, and 4 correctly describe data wrangling. Statement 2 describes a data modeling process.

**Question 2 (Spend 2.13).** A company conducts a survey of 500 consumers. The accompanying data file shows their home ownership (OwnHome: Yes/No), car ownership (OwnCar: Yes/No), annual household spending on food (Food), and annual household spending on travel (Travel).

a. Sort the data by home ownership, car ownership, and the travel spending all in descending order. How much did the first customer on the ordered list spend on food?
b. Sort the data only by the travel spending amount in descending order. Of the 10 customers who spend the most on traveling, how many of them are homeowners? How many of them are both homeowners and car owners?
c. How many missing values are there in each variable?
d. How many customers are homeowners?
e. How many customers are homeowners but do not own a car?

**Answer 2.**

a. 3362.86
b. Only 3 of them are homeowners, 1 has both a home and car owner.

c. 3 values are missing for the variable OwnHome, 3 values are missing for the variable OwnCar, 0 Values are missing for FoodSpend, 1 Values is missing for TravelSpend.
d. 133
e. 88

**Question 3 (Salaries 2.30).** Ian Stevens is a human resource analyst working for the city of Seattle. He is performing a compensation analysis of city employees. The accompanying data set contains three variables: Department, Job Title, and Hourly Rate (in $). A few hourly rates are missing in the data.

a. Split the data set into a number of subsets based on Department. How many subsets are created?
b. Which subset contains missing values? How many missing values are in that data set?
c. Use the imputation strategy to replace the missing values with the mean of the variable. What is the average hourly rate for each subset?

**Answer 3.**

a. 4 subsets created
b. Subset IT Dept has 2 missing values.
c.

| IT | HR | PU | SE |
|---|---|---|---|
| 52.1505 | 47.02796 | 41.15157 | 46.84691 |

**Question 4 (Exercise_2.42).** The accompanying data file contains two variables, Date1 and Date2.

a. Create a new variable called "DifferenceInYear" that contains the difference between Date1 and Date2 in year for each observation. What is the average difference in year? (Hint: Use the YEARFRAC function if you are using Excel to complete this problem.)
b. Create a new variable called "Month" that contains the month values extracted from Date1. What is the average month value?
c. Bin the month values in four equal-interval groups. Label the groups using numbers 1 (lowest values) to four (highest values). Which group has the highest number of observations?

**Answer 4.**

a. 29.00791667
b. 6 (June)
d. Group(3) has the highest number of observations.

**Question 5 (Exercise 2.49).** The given table has three variables and six observations.

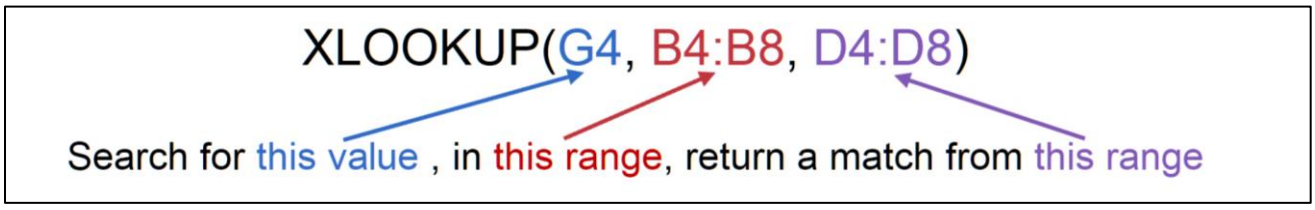
a. Convert Sex into dummy variables. Use the most frequent category as the reference category. Which category is the reference category?
b. Transform Decision into dummy variables. Use the most frequent category as the reference category. Which category is the reference category?
c. Transform the Decision values into category scores where Approve = 1, Reject = 2, and Need More Information = 3. How many observations have a category score of 2?

**Answer 5.**

a. Male is the reference category.
b. Approve is the reference category.
c. 1

## Part 2 - Extra Practice Questions for Lookup Functions

- **VLOOKUP**: You can use the VLOOKUP function to search the first column of a range (range: Two or more cells on a sheet. The cells in a range can be adjacent or nonadjacent.) of cells, and then return a value from any cell on the same row of the range
- **HLOOKUP**: Searches for a value in the top row of a table or an array of values and then returns a value in the same column from a row you specify in the table or array.
    - Use HLOOKUP when your comparison values are in a row across the top of a table of data, and you want to look down at a specified number of rows.
    - Use VLOOKUP when your comparison values are in a column to the left of the data you want to find.
    - The V in VLOOKUP stands for "Vertical" and The H in HLOOKUP stands for "Horizontal"
- **XLOOKUP:** works on multiple values. You can combine XLOOKUP with additional functions for more complex lookups.

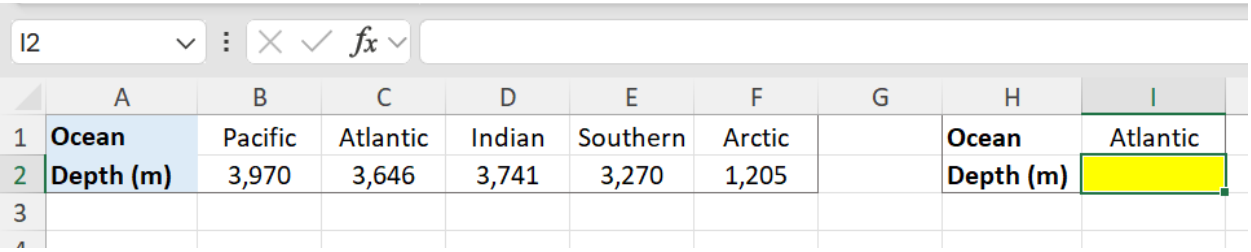


1. Get the name of each student's department using the department code.

| Name | Department code | Department Name |
|---|---|---|
| Adella | DS | |
| Allen | CS | |
| Anne | CS | |
| Beverly | IT | |
| Bonnie | DS | |
| Brian | IS | |
| Carol | DS | |
| Christopher | IT | |
| Coral | IS | |
| Daryl | CS | |

| Department code | Department Name |
|---|---|
| DS | Decision Support |
| CS | Computer Science |
| IT | Information Technology |
| IS | Information Systems |

2. Find the depth of a specific ocean. Remember: for h-lookup, enter rows instead of columns.




3. Get **both** the Math and English marks for Carry. Perform a two-way XLOOKUP.

